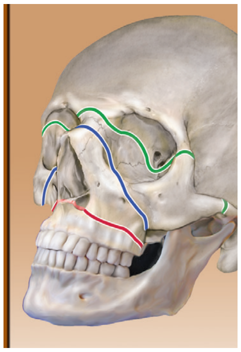

25

Sports Nutrition

Jessica L. Buschmann, Jackie Buell

Proper nutrition is often an undervalued facet of athletic competition. However, in recent years sports nutrition has gained popularity and has secured recognition as a key piece of the training regimen. Within the past decade there has been a notable increase in sports nutrition original research publications. It is undoubtedly an emerging area of science and practice that continues to thrive and progress. Proper fueling techniques equip an athlete with adequate energy to train and compete, enhance recovery post workout, manage optimal body composition, prevent and/or facilitate injury healing, strengthen the immune system, and reduce or delay the onset of fatigue. Athletes interested in seeking an individualized nutrition plan are encouraged to seek a Board Certified Specialist in Sports Dietetics (RDN, CSSD) by visiting the Sports, Cardiovascular, and Wellness Nutrition Practice Group (SCAN) website: http://www.scandpg.org/search-rd/.[1]

ENERGY BALANCE

Energy balance is the primary nutritional consideration for athletes' health and performance, especially for those trying to manipulate body weight (BW) and/or composition. Losing BW requires an athlete to eat fewer calories than expended, while gaining weight requires a small overconsumption of calories. Intentional energy imbalance should be within 300 to 500 calories of the energy needs to avoid losing lean mass in weight loss programs[2] or gaining fat in weight gain programs (unless the athlete is indifferent to the composition of the weight gain). Getting the energy balance fine-tuned to help the athlete reach performance and body goals is likely the trickiest piece of nutritional counseling.[3] Equations are often used to estimate resting metabolic rate, and then are adjusted for the amount of physical activity.[4] When using generic values, the growth status, body size, and volume and intensity of training will influence the predicted energy needs. In general, female athletes of an average body size will need between 2000 and 2400 calories per day. Growing male athletes may need upwards of 3500 to 4500 calories, while average size adult males (such as collegiate athletes) may be more reasonable at 3200 to 3600 calories. The current paradigm used to look at energy balance in athletes is the energy availability (EA) equation, which requires the practitioner to know both the body composition and the energy expenditure of exercise.[5] The ideal EA for athletes is greater than 45 kcal/kg lean mass while less than 30 is currently the definition of inadequate EA.[2] Low EA is the likely genesis of the Female Athlete Triad[6] and is now part of a larger energy insufficiency paradigm, coined "relative energy deficiency in sport" (RED-s).[2] Again, there are a lot of factors to consider in the estimation of energy needs, and this is a good space for a sports dietitian to help your practice.

MACRONUTRIENTS: CARBOHYDRATE, PROTEIN, AND FAT

The macronutrient profile of the performance diet should also match the athlete's sport, position or event, and the desired body changes. Prolonged high-intensity exercise requires carbohydrate (CHO) as the primary fuel for muscle contraction, and glycogen is the storage form of CHO in the muscle.[4] Athletes who require "high gear" should consume a diet adequate in CHO to keep muscle glycogen reserves maximized in preparation for performance.[2] The literature also suggests that CHO adequacy will support the immune system and help athletes avoid overtraining syndrome.[3] CHO needs can vary widely by sport, intensity, and volume of exercise (Table 25.1). Dairy products, fruits, and grains are the primary CHO sources for the training diet.

The defined protein needs for athletes has steadily increased with additional research, and current recommendations may be justified as high as 2.2 g/kg BW to optimize lean mass accretion in male resistance-trained athletes.[7] Most athletes would benefit from a protein intake in the range of 1.2 to 1.7 g/kg BW.[3] While it is suggested in the literature that males need more protein than females, there is no demonstrated harm for slight overconsumption of protein for an athlete with healthy liver and kidneys, and the extra protein may benefit body composition.[8] Optimizing protein intake is important to ensure muscle recovery and training adaptations as well as hormonal health. Meats, dairy, and beans are among the good protein sources for the training diet.

How much fat to feed athletes is a source of constant contention among practitioners, and is typically expressed as a percentage of calories in the training diet. Within an eating pattern where all food groups are included, most sports nutrition guidelines agree that 20% to 35% of the calories should come from a variety of fat-containing foods for good health and muscle performance.[3] Athletes restricting CHO as part of a well-formulated ketogenic or metabolic optimizing program will likely consume a much higher amount of dietary fat.[10] Fats have long been held as the culprit in heart disease and obesity, and this has led to fear of consuming fats. Athletes need to understand

Abstract

Proper nutrition is often an undervalued facet of athletic competition. However, in recent years sports nutrition has gained popularity and secured recognition as a key piece of the training regimen. It is undoubtedly an emerging area of science and practice that continues to thrive and progress. Proper fueling techniques equip an athlete with adequate energy to train and compete, enhance recovery post workout, manage optimal body composition, prevent and/or facilitate injury healing, strengthen the immune system, and reduce or delay the onset of fatigue. The goal of this chapter is to provide practical recommendations for sports medicine professionals for the elite and recreation athlete alike. Within this chapter, energy balance, role of macronutrients, vitamins/minerals (iron, calcium, iron) hydration strategies, nutrient timing, and identifying reliable resources for third-party verification of sports supplements will be discussed.

Keywords

macronutrients
hydration
calcium
iron
vitamin D
nutrient timing
energy balance

TABLE 25.1 Macronutrient Needs for Athletes

	Endurance	Strength	High Intensity	Recreational
CHO needs (g/kg/day)	6–12	3–7	5–7	3–5
Protein needs (g/kg/day)	1.2–1.7	1.6–2.4	1.4–2.2	1.2–1.7
Special population concerns	Bone health with low EA Iron status	Third-party approval of PES if used	Pre-event and recovery fueling important	Over-fueling leads to weight gain

Endurance: Sports or events performed at submaximal effort over longer periods of time.
Strength: Athletes trying to increase lean mass and strength in a periodized program.
High Intensity: Sports that require multiple bursts of high-intensity performance (most team sports).
Recreational: Usual workout is less than 1 hour once per day, and performance not a priority.
CHO, carbohydrate; *EA*, energy availability; *PES*, performance-enhancing supplements.
Values have been adapted from references 9, 27, and 29 position statements.

that periods of low intensity exercise rely on fat oxidation[4] and the intramuscular triglyceride (IMTG) pool is an important source of muscle energy.[11] Foods such as oils, nuts/seeds, avocados, and salad dressings should be included in a balanced diet.

Appropriate to age, the practitioner should remember to query the athlete on the nonessential macronutrient: alcohol. Athletes often use alcohol for relaxation and social enjoyment, and alcohol can decrease performance as well as negatively impact body composition and increase calorie intake. Alcohol after exercise also impedes glycogen recovery.[12] The intentional macronutrient composition of the athlete's diet can make or break their performance and health goals. Table 25.1 provides the current sports nutrition macronutrient guidelines adjusted for the type of sport.

HYDRATION

Adequate hydration practices are critical for optimal performance and safety standards for athletes. Water makes up approximately 70% of tissue—including muscle tissue—within the body. Adequate hydration helps maintain countless functions within the human body, including food absorption, digestion, transporting nutrients, body temperature regulation, joint lubrication, waste removal, and the protection of organs. The hydration needs of athletes vary greatly based on individual preference, sweat rate, acclimation to the environment, characteristics of the sport, and the duration and intensity of exercise.[13] Three common concerns surrounding hydration status include dehydration, exercise-associated hyponatremia (EAH), and exercise associated skeletal muscle cramps (EAMC). If athletes drink when thirsty, have access to free clean water, and implement a hydration plan, he or she should be able to maintain a euhydrated state.[13,14]

Hydration Considerations

Many athletes struggle with under-hydrating when preparing for or participating in competitions. A key consequence of dehydration is an increase in the body's core temperature during physical activity, causing impaired skin blood flow and altered sweat responses. Body temperature has the ability to rise 0.15°C to 0.20°C for every 1% weight lost secondary to sweating during physical activity.[13] Early signs of dehydration include thirst, general discomfort and complaints; more severe signs soon follow, including flushed skin, cramps, dizziness, headache, vomiting, nausea, heat sensations on the head or neck, chills, decreased performance, and dyspnea.

Conversely, the risk for overhydration should not be underestimated. EAH is commonly observed in recreational or novice athletes participating in endurance events such as half marathons, sprint triathlons, and military training events. EAH commonly stems from overconsumption of water or other hypotonic beverages in events lasting more than 4 hours. EAH is defined by a blood sodium concentration below the normal reference range, which for most laboratories is a value below 135 mmol/L.[15] Signs and symptoms of EAH may occur during or up to 24 hours after physical activity and may include vomiting, headache, confusion, delirium, seizures, respiratory distress, and loss of consciousness. Without immediate medical attention, EAH can result in severe consequences, including death.

Despite the commonality of muscle cramping with exercise, etiology is often multifactorial and overall is not well understood. Despite popular belief, dehydration and electrolyte imbalances are not evidence-based causes of EAMC.[16–18] Strong mounting evidence shows that EAMC is likely linked to a neuromuscular control abnormality and muscle fatigue. Athletes prone to cramping are advised to focus on sound nutrition principles, regular stretching routines, appropriate hydration, and the prevention of muscle damage by tapering and maintaining proper conditioning.

Monitoring Hydration

Hydration status can be monitored in several ways, such as observing weight changes pre/post exercise, subjective feelings, urine color, and thirst sensation. Studies show maintaining a hydration schedule versus ad libitum drinking helps maintain and/or improve performance. However, it is advisable for athletes to listen to their bodies and adjust the plan if needed, particularly if an individual experiences significant weight fluctuations during exercise.[14] If an athlete has historic concern for maintaining weight during exercise, thirst alone may not be enough to promote a euhydrated state.[19] Current guidelines suggest limiting loss of fluids to less than 2% of total BW compared to their pre-competition weight. Athletes with an observed fluid loss greater than 2% may experience impaired cognitive function and exercise performance, especially in hot weather.[13,14] Despite this recommendation, a recent study observed that ultra-endurance runners with greater than 3% weight loss did not experience an adverse effect during their performance.[20] In addition, athletes with a high sweat rate and elevated sweat sodium concentration ("salty sweater") present a greater risk for developing muscle cramps secondary to electrolyte imbalances and hypohydration.[14] Athletes not acclimatized to heat and the surrounding environment are particularly susceptible.[21] Therefore a calculated sweat rate for athletes who have challenges with hydration, have exercise-associated muscle cramps, or who perform at an elite level is encouraged. While there is weak evidence to support formal recommendations, Table 25.2 offers suggestions for the timing of fluid intake before, during, and after exercise.

VITAMINS AND MINERALS

Many athletes chose to take a "more is better" approach to supplementation versus ensuring consumption of a diet rich in a variety of wholesome, natural food. It is well recognized that diets low in certain vitamins and minerals may lead to health and performance side effects. However, athletes who consume above the tolerable upper intake level limit may also result in undesired adverse reactions. Athletes are therefore encouraged to focus on consuming adequate amounts of vitamins and minerals, which is ideal for absorption and metabolic functions throughout the body.

Iron

Iron plays a critical role in helping athletes achieve peak performance, namely in terms of aerobic endurance and muscular function. Iron is responsible for oxygen transport, energy metabolism, erythropoiesis (red blood cell production), immune system maintenance, and thyroid hormone function.[4] Hemoglobin and myoglobin bind oxygen by means of the porphyrin ring of heme.[22] Hemoglobin in red blood cells carries oxygen to the exercising skeletal muscles, whereas myoglobin is responsible for oxygen transfer from erythrocytes to muscle cells—more specifically, the mitochondria. Within the mitochondria, iron is a part of the cytochrome system for oxidative phosphorylation, along with other oxidative enzymes responsible for the generation of energy available to the body.[4] Therefore an athlete with iron deficiency will experience reduced endurance capacity secondary to impaired ATP production.[22] Low iron stores can result from insufficient consumption of iron-rich food sources, periods of rapid growth, training at high altitudes, menstrual blood loss, foot-strike hemolysis, blood donation, or injury.[23] Athletes at the greatest risk for developing low iron levels include females, distance runners, and plant-based athletes.

Dietary sources of iron vary based on type. Heme iron is found in animal sources including red meat, pork, seafood, and dark cuts of poultry. Nonheme iron sources are plant-based and are not absorbed as readily compared to their heme iron counterparts. The absorption of heme iron is enhanced when consumed alongside a vitamin C source such as dark green leafy vegetables, peppers, broccoli, melons, tomatoes, berries, and citrus fruits. Conversely, several food items are known to lead to impaired absorption of iron including calcium-enriched products, oxalates (spinach, beets, nuts, fresh herbs), polyphenols (black tea, coffee, cocoa, berries), and phytates (soy protein, fiber, lentils, whole grains). If an athlete intends to ingest a food item containing one of the above, it is recommended to delay consumption of an iron-rich food item for 2 hours.[24]

Calcium

Calcium is an important mineral required for bone health including growth, maintenance, repair, and structure. In addition, calcium is also required for muscle contraction, nerve transmission, and blood clotting. Deficiency can lead to weak bones, increased risk for fracture or stress fracture, and blood pressure disturbances. Dietary sources of calcium include dairy products, leafy green vegetables (when cooked), fortified food items (juice, cereal), alternative milks (soy, almond), and legumes. Leafy greens can be cooked (typically blanched) in order to reduce the amount of oxalate in the vegetable, therefore improving the delivery potential for calcium to the body.

Vitamin D

Vitamin D has a significant impact on maintenance of adequate bone density, muscle strength, immune system function, possession of antiinflammatory properties, and hormonal regulation. Vitamin D2 is found in some plants consumed in the diet; it is produced commercially by the irradiation of yeast, and is fortified in some food products. Vitamin D3 (the active form) is manufactured in the skin in response to ultraviolet B (UBV) rays from sunlight; additionally, dietary or animal sources, such as deep sea fatty fish, egg yolks, or liver, also can provide this vitamin. Supplements of Vitamin D3 also may be taken when needed for plant-based athletes given food sources that are animal based.

Another potential benefit of vitamin D is linked to its potential antiinflammatory properties. Intense exercise, overtraining, and

TABLE 25.2 Hydration Timing Techniques

Exercise	Fluid Goals	Timing	Additional Goals
Pre-exercise	17–20 ounces 7–10 ounces	2–3 h pre-exercise 10–20 min pre-exercise	• Have a little salt with meal or snack to promote water retention
During exercise	7–10 ounces	Every 10–20 min during activity if able or tolerated	• Be mindful of acceptable individual preferences and sweat rates • Sweat rates vary amongst athletes • Find a sports drink palatable to the athlete
Postexercise	19–23 ounces fluid per pound lost	After exercise; ideally within 2 h post exercise	• Replace fluid losses completely especially if going into another exercise session • Discourage alcohol intake

Values adapted from References 13 and 14.

other sports-related injuries are known to elevate levels of pro-inflammatory cytokines. Vitamin D has been shown to reduce the production of these cytokines while increasing the production of antiinflammatory cytokines.[25] This suggests that an adequate supply of vitamin D may reduce inflammation and muscle soreness, and promote quicker recovery after intense exercise. Athletes with a history of stress fracture, bone or joint injury, signs of overtraining, muscle pain or weakness, and a lifestyle involving low exposure to UVB rays may benefit from lab work to determine if an individualized vitamin D supplementation protocol is required.[21]

NUTRIENT TIMING

Proper nutrient timing should be a fundamental part of an athlete's fueling plan. A well-balanced nutrition plan should not only contribute to performance enhancement, but also to an athlete's psychological and gastrointestinal comfort. Avoidance of dehydration, electrolyte imbalances, glycogen depletion, hypoglycemia, and acid-base disturbances are also notable considerations.[32] Nutrient timing and strategies vary greatly based on duration of activity, type of sport, and time between preceding or upcoming events. Food availability and an athlete's personal food preferences are also important considerations. Table 25.3 provides basic strategies to help an athlete fuel before, during, and after exercise.

SPORTS SUPPLEMENTATION

Many athletes feel the need to use dietary supplements. It is important to emphasize to an athlete that these "Performance Enhancing Supplements" (PES) are meant to *supplement* a strong training diet. Dietary supplements are poorly regulated and athletes consuming tainted products are at risk for disqualification regardless of how the illegal ingredient got into their body.[33] This must be taught and emphasized to high-level athletes from a young age to avoid mishaps leading to the stripping of a coveted championship and a tarnished reputation. The National Collegiate Athletic Association, International Olympic Committee (IOC), and other sports organizations and federations usually provide a published list of banned substances with a championship drug testing process to ensure a level playing field. Researching the safety, purity, and efficacy of products reveals a complex and daunting process.[33] Table 25.4 provides current commonly consumed PES with the purported benefits and usual protocols. This information is provided for professional guidance, not to encourage use.

For a complete list of references, go to ExpertConsult.com.

TABLE 25.3 Nutrient Timing Strategies

<table>
<tr><th>Timing Recommendations</th><th>Goals</th></tr>
<tr><td>Pre-exercise
>60 min before activity</td><td>
Timing, amount and type of food and drink should match the practical needs of the event and the individual preferences
Foods higher in fat, protein, or fiber may cause gastrointestinal issues during activity
Carbohydrate goals:
Light: low intensity or skilled-based activity
3–5 g/kg/day
Moderate: 1 h exercise program per day
Carbohydrate 5–7 g/kg/day
High: 1–3 h/day moderate/high intensity
6–10 g/kg/day
Very high: extreme >4–5 h/day moderate/high intensity
8–12 g/kg/day

</td></tr>
<tr><td>During exercise</td><td>
Brief: <45 min
Nothing needed
Prolonged high intensity: 45–75 min
Small amounts if needed per individual preference
Endurance: 1–2.5 h
30–60 g carbohydrate
Example: 8 oz. sports drink
Example: (3) Clif Shot Bloks
Ultra-endurance: >2.5 h
Up to 90 g of carbohydrate per hour
</td></tr>
<tr><td>Post exercise
Aim for 45–60 min after activity</td><td>
Focus on carbohydrate and protein food pairings
Carbohydrate: Replenishing glycogen stores and limiting cortisol-induced muscle damage during exercise
Recommendation: 1 g/kg
Protein: crucial to muscle protein synthesis and repair
Recommendation: 20 g of high-quality protein

Meal examples:
Example: 8 oz. chocolate milk with turkey sandwich (2 slices of whole grain bread & 3 oz. turkey)
Example: 1 cup ready-to-eat cereal (i.e., Cheerios) with 1 cup skim milk and 1 medium banana

</td></tr>
</table>

Values adapted from reference 27, 29, 31, and 32.

TABLE 25.4 Common Performance Enhancing Substances

Substance	Purported Benefit	Usual Dose and Timing to Event	Comments
Creatine	Increased muscle ATP allowing higher workload	3–5 g/day, not necessarily important when taken[33]	Muscle will hold more water and increase weight of athlete
Nitrates (e.g., beetroot juice)	Precursor to nitric oxide (vasodilator) thus improving tissue oxygenation and metabolism	0.1 mmol/kg/day or 5–6 mmol/day over 5–6 days or more[34]	Complex pathway of conversion to nitric oxide in the body
Beta alanine	Precursor to muscle carnosine thus improves muscle buffering allowing higher workload	Small doses multiple times per day, 4–6 g/day in total for at least 4 weeks[7]	May cause tingling or paresthesia (thought to be harmless)
Caffeine	Decreases perception of fatigue centrally, or peripheral mechanism	3–6 mg/kg 1 h prior to event 1–2 mg/kg during later endurance tasks[33]	
Pre-workouts	Help athlete increase arousal	No set research numbers	Ensure third-party verification to help avoid illegal ingredients such as stimulants
Mass gainers	Help athlete gain lean mass	No set research numbers	Consider the extra calories in overall diet; ensure third-party verification to help avoid illegal ingredients such as anabolic agents

SELECTED READINGS

Citation:
American College of Sports M, Sawka MN, Burke LM, et al. American College of Sports Medicine position stand. Exercise and fluid replacement. *Med Sci Sports Exerc*. 2007;39(2):377–390.

Level of Evidence:
Various topics defined by position statement

Summary:
The American College of Sports Medicine on exercise and fluid replacement for athletes provides a review of current recommendations and strategies to optimize fluid-replacement practices of athletes.

Citation:
Thomas DT, Erdman KA, Burke LM. Position of the Academy of Nutrition and Dietetics, Dietitians of Canada, and the American College of Sports Medicine: Nutrition and Athletic Performance. *J Acad Nutr Diet*. 2016;116(3):501–528.

Level of Evidence:
Various topics defined by position statement

Summary:
The Academy of Nutrition and Dietetics, Dietitians of Canada, and the American College of Sports Medicine merged to create a Position Statement regarding Nutrition and Athletic Performance. These organizations provide guidelines for the appropriate type, amount, and timing of intake of food, fluids, and supplements to promote optimal health and performance across different scenarios of training and competitive sport.

Citation:
Buell JL, Franks R, Ransone J, et al. National Athletic Trainers' Association. National athletic trainers' association position statement: evaluation of dietary supplements for performance nutrition. *J Athl Train*. 2013;48(1).

Level of Evidence:
Various topics defined by position statement

Summary:
The National Athletic Trainer's Position Statement on the evaluation of dietary supplements for performance nutrition promoting a "food-first" philosophy to support health and performance; to understand federal and sport governing body rules and regulations regarding dietary supplements and banned substances; and to become familiar with reliable resources for evaluating the safety, purity, and efficacy of dietary supplements.

Citation:
Meeusen R, Duclos M, Foster C, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. *Med Sci Sports Exerc*. 2013;45(1):186–205.

Level of Evidence:
Various topics defined by position statement

Summary
Within this consensus statement between the American College of Sports Medicine and European College of Sport Science the current state of knowledge on overtraining syndrome (OTS) including definition, diagnosis, treatment and prevention is discussed.

Citation:
Mountjoy M, Sundgot-Borgen J, Burke L, et al. The IOC consensus statement: beyond the Female Athlete Triad–Relative Energy Deficiency in Sport (RED-S). *Br J Sports Med*. 2014;48(7):491–497.

Level of Evidence:
Various topics defined by position statement

Summary:
This Consensus Statement replaces the previous and provides guidelines to guide risk assessment, treatment, and return-to-play decisions. The IOC expert working group introduces a broader, more comprehensive term for the condition previously known as *Female Athlete Triad*. This Consensus Statement also recommends

practical clinical models for the management of affected athletes. The "Sport Risk Assessment and Return to Play Model" categorizes the syndrome into three groups and translates these classifications into clinical recommendations.

Citation:

IOC consensus statement on sports nutrition 2010. *J Sports Sci*. 2011;29(suppl 1):S3–S4.

Level of Evidence:

Various topics defined by position statement

Summary:

A practical guide to eating for health and performance prepared by the Nutrition Working Group of the International Olympic Committee.

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